

Domain-Generalized Road Crack Segmentation via Causal-Inspired Channel Attention

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Abstract—This paper introduces a causality-inspired channel attention module to enhance the domain generalization of crack segmentation networks under diverse environmental conditions. The proposed module selectively normalizes feature responses sensitive to confounding factors, while preserving channel responses that are more likely to encode structurally relevant and causally stable features, thereby improving the generalization of crack representations. To better identify confounder-sensitive channels that are sensitive to confounding factors, an alternating intervention-driven learning strategy is developed. Experimental results on public datasets demonstrate the effectiveness of the proposed method in improving model generalization. Moreover, by integrating the module with a lightweight backbone and deploying it on an embedded device mounted on a self-built mobile robot, the system achieves real-time road crack segmentation at 22.7 FPS, validating its practical applicability.

Keywords—causal mechanism, channel attention module, road crack segmentation, mobile robot

I. INTRODUCTION

Transportation infrastructure is vital for public welfare and economic development. As pavements age, cracking becomes a prevalent issue, compromising structural integrity and posing safety risks. Accurate crack detection is therefore essential for road maintenance.

Recent advances in deep learning have made vision-based crack detection an active research area. Semantic segmentation networks such as U-Net outperform traditional methods on various datasets. However, their robustness in real-world

scenarios remains challenging due to illumination changes, shadows, and varying pavement textures—causing significant performance drops across different deployment environments.

This robustness issue stems from limited domain generalization: models rely on spurious correlations (e.g., lighting or background) rather than causal crack features. Prior solutions, such as data augmentation or domain-adversarial learning do not address this fundamental limitation.

We approach this problem from a causal representation learning perspective. Building on the lightweight RHACrack-Net, we introduce a causal-inspired module with instance-wise attention to suppress feature responses sensitive to environmental confounders, while enhancing the stability of detecting crack-related causal components—without modifying backbone architecture or adding inference overhead. Experiments on benchmark datasets and counterfactual evaluations demonstrate improved generalization under illumination and texture variations.

II. RELATED WORK

A. Frequency-based Domain Generalization

Frequency-domain transformations have been shown effective for mitigating domain shifts by separating structural and style information. FDA [1] swaps amplitude spectra between images via FFT, achieving strong performance in domain-generalized semantic segmentation. FACT [2] further improves generalization by mixing amplitude spectra across domains. More recently, frequency-based representations have been explored in crack segmentation; for example, CDE-Crack [3] fuses amplitude spectra while preserving crack geometry through shape-aware stylization.

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However, long and thin cracks rely heavily on high-frequency edge information while being vulnerable to low-frequency illumination and texture variations. Pure frequency-based approaches therefore may degrade fine-grained segmentation under severe domain shifts. Instead of relying solely on frequency manipulation, we investigate the relationship between frequency-domain representations and causal factors, aiming to enhance generalization by suppressing environment-dependent components.

B. Causal Mechanism in Domain Generalization

Causal learning provides an alternative perspective for domain generalization by identifying invariant causal features while reducing the influence of spurious correlations. Lv et al. [4] extract causal representations from inputs to improve robustness across domains. Zhang et al. [5] introduce a causal intervention module to learn environment-invariant features under adverse conditions. UFR [6] further constructs a structural causal model and proposes a causal attention mechanism to mitigate domain bias.

In crack detection, existing methods such as CrackUDA [7] mainly rely on adversarial alignment or style transfer, which may still capture pavement textures and illumination as spurious cues. In contrast, the proposed method explicitly modulates feature responses using a plug-and-play Causal Attention Mechanism (CAM) guided by intervention consistency, enabling lightweight and stable generalization for fine crack structures.

III. METHOD

To address cross-domain generalization in complex open environments, we adopt a causal perspective. Robust crack segmentation relies on suppressing environment-sensitive confounding factors while preserving stable, structure-related representations. We therefore introduce a lightweight, plug-and-play, causal-inspired *Causal Attention Mechanism* (CAM) and an *alternating intervention-driven training strategy* that are integrated into RHACrackNet. Fig. 1 illustrates the overall framework of the proposed method and the detailed structure of CAM.

Structurally, CAM is inserted into the four decoder blocks of our RHACrackNet backbone (a ResNet-augmented U-Net with augmented attention on skip connections and bottleneck). CAM directly replaces the original batch-normalization layers in the decoder, enabling instance-level, sample-dependent modulation after semantic fusion. At the optimization level, controlled environmental interventions and consistency constraints guide CAM to focus on crack-geometry-related invariant features.

A. Causal Attention Mechanism

Crack features in an image $F \in \mathbb{R}^{C \times H \times W}$ comprise two latent components: (i) stable geometric cues (edges, connectivity) and (ii) environment-sensitive statistics (illumination, shadows, texture). Based on per-sample channel statistics, CAM implicitly modulates these components, preserving

structure-related representations while suppressing responses sensitive to environmental variations.

For each sample, a spatially aggregated response is computed for each channel.

$$z_c = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W F_c(i, j), \quad c = 1, \dots, C. \quad (1)$$

GAP extracts channel-wise statistics, which are then processed by an MLP to model inter-channel relationships and generate attention weights.

$$\alpha = \sigma(\text{MLP}(z)) \in [0, 1]^C. \quad (2)$$

We then adaptively modulate each channel by

$$F'_c = \alpha_c F_c + (1 - \alpha_c) \tilde{F}_c, \quad (3)$$

where $\tilde{F}_c$ denotes the statistically balanced response obtained via instance normalization.

$$\tilde{F}_c = \frac{F_c - z_c}{\sqrt{\sigma_c^2 + \varepsilon}}, \quad (4)$$

where σ_c^2 denotes the spatial variance of channel c , and $\varepsilon = 10^{-5}$ is used for numerical stability.

When $\alpha_c \approx 1$, the original structure-rich response is preserved; when $\alpha_c \approx 0$, the balanced term suppresses environment-induced shifts. Because the decoder progressively reconstructs crack topology, placing CAM at the BN positions after skip-attention fusion effectively attenuates confounding statistics while retaining fine geometric detail.

B. Alternating Intervention-Driven Training

We introduce an intervention-driven strategy that adaptively optimizes the module parameters by enforcing consistency constraints under perturbations, encouraging the attention weights to focus on task-relevant features with causality-inspired stability while mitigating spurious correlations. We model an image as $x = g(s, e)$ where s denotes crack structure and e environmental factors, and the label $y = f(s)$ depends only on s . During training we generate intervened samples $x' = g(s, e')$ via four lightweight, structure-preserving perturbations—frequency-domain illumination redistribution, long-edge shadow injection, texture amplitude-spectrum mixing, and Gaussian blur—which simulate real-world variations in lighting, shadow, texture, and focus while leaving the underlying crack geometry unchanged.

Prediction consistency on intervened samples is enforced by

$$\mathcal{L}_{\text{cons}} = \|f(x) - f(x')\| \quad (5)$$

and the overall loss for an intervened batch is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}}(x') + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}} + \lambda_{\text{anchor}} \mathcal{L}_{\text{anchor}}, \quad (6)$$

where $\mathcal{L}_{\text{seg}}$ is the standard combination of binary cross-entropy and Dice losses, and $\mathcal{L}_{\text{anchor}}$ is the segmentation loss evaluated on the original (clean) sample only during the intervention stage to serve as an anchor preventing excessive suppression of crack-related structural channels. To avoid over-regularization, we adopt an alternating optimization strategy

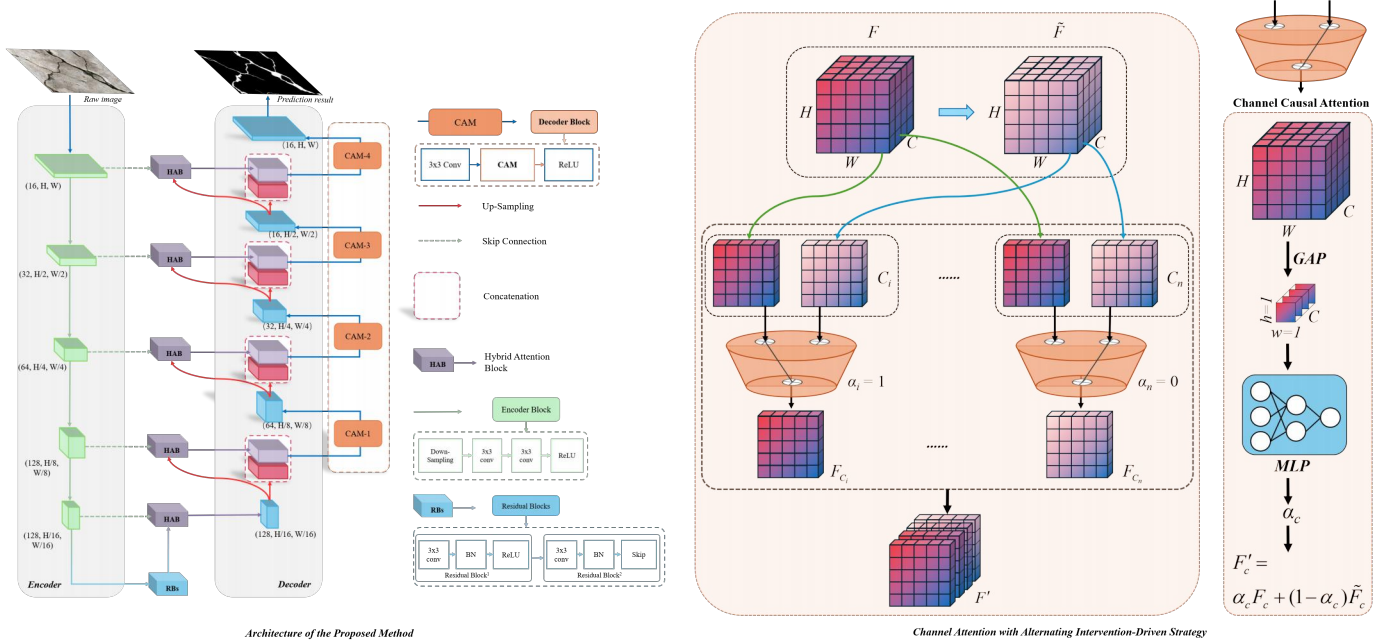


Fig. 1. Overview of the proposed framework and detailed architecture of the CAM.

within each training epoch: (1) a structure-dominant stage that updates only the backbone parameters on clean samples using $\mathcal{L}_{\text{seg}}(x)$; and (2) an intervention-driven stage that updates the CAM (together with light updates to the backbone) on intervened samples using $\mathcal{L}_{\text{total}}$.

This alternating schedule lets the backbone learn robust structural representations while CAM absorbs and suppresses environment-induced fluctuations. Consequently, channels that stably encode task-causal components receive persistently high α_c across domains, and channels sensitive to non-causal environmental components are gradually down-weighted, yielding superior cross-domain generalization without architectural overhead or extra inference cost.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

For evaluation, we integrate the proposed CAM into the RHACrackNet [8] baseline by replacing the batch normalization layers in its decoder, keeping the backbone unchanged to ensure fair comparison. We use three public datasets: CamCrack789 [8] (789 concrete crack images) serves as the source domain, split into training (70%), validation (10%), and test (20%) sets. To assess cross-domain generalization under texture shifts, we directly test models trained on CamCrack789 on Crack500 [9] (500 asphalt crack images) and CFD [10] (118 concrete crack images). All models are trained with the same protocol: AdamW optimizer (lr=0.001, weight decay=5e-4), batch size 4, on an NVIDIA RTX 3090. We evaluate the models using four widely used semantic segmentation metrics [11]: precision, recall, F1, and IoU.

B. Performance under Controlled Environmental Interventions

To assess robustness under environmental distribution shifts, we introduce controlled environmental interventions on the CamCrack789 test set while preserving crack structures. Five perturbation types are considered: illumination, shadow, texture, blur, and random combinations of these disturbances. All perturbations are applied to the original test images without altering annotations, ensuring structural consistency.

TABLE I
PERFORMANCE UNDER CONTROLLED ENVIRONMENTAL INTERVENTIONS

Perturbation	Model	Precision	Recall	F1-score	IoU
Raw	Baseline	0.9191	0.9184	0.9187	0.8497
	Ours	0.9173	0.9280	0.9226	0.8564
Combinations	Baseline	0.8410	0.7870	0.8131	0.6851
	Ours	0.8620	0.8918	0.8767	0.7804
Illumination	Baseline	0.8266	0.7724	0.7986	0.6647
	Ours	0.8687	0.9066	0.8872	0.7974
Texture	Baseline	0.8603	0.8022	0.8302	0.7098
	Ours	0.8754	0.8274	0.8508	0.7403
Blur	Baseline	0.8782	0.8362	0.8567	0.7493
	Ours	0.9002	0.8707	0.8852	0.7940
Shadow	Baseline	0.8296	0.7042	0.7618	0.6152
	Ours	0.8958	0.9076	0.9017	0.8210

As shown in Table I, the proposed method consistently outperforms the baseline across all perturbations, with the largest gains under severe conditions such as shadow and combined disturbances. For example, under shadow perturbation, F1 improves from 0.7618 to 0.9017 and IoU from 0.6152 to 0.8210, indicating strong robustness to illumination-related artifacts commonly encountered in real road scenes.

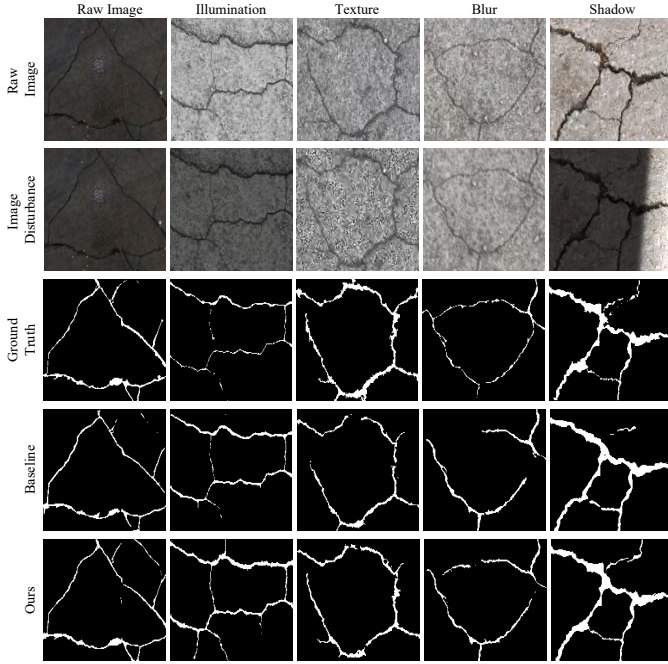


Fig. 2. Visual comparison of detection results under different environmental perturbations.

Fig. 2 provides qualitative comparisons of segmentation results under different perturbations. While the baseline model exhibits missed detections and fragmented predictions when environmental conditions change, the proposed method maintains more complete and continuous crack structures, demonstrating improved stability across varying visual appearances.

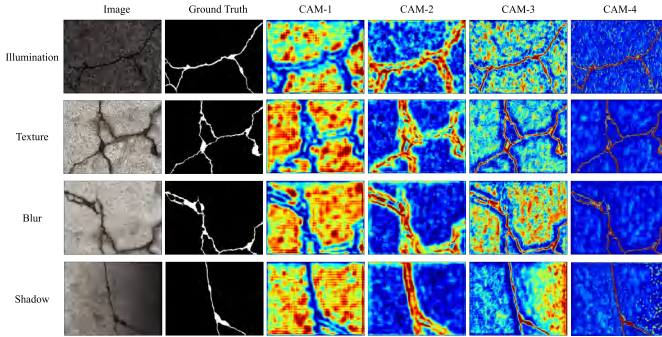


Fig. 3. Activation heatmaps of CAM at different decoder stages.

To further analyze model responses, Fig. 3 visualizes the activation maps of the four inserted CAM modules. Despite significant variations in brightness, texture, blur, and shadow, the high-response regions remain concentrated along crack structures across different layers, while background responses are largely suppressed. This observation suggests that the model tends to emphasize structure-related features while reducing sensitivity to environment-dependent patterns.

Overall, the results demonstrate that the proposed channel attention mechanism promotes reliance on structure-related

features, leading to more stable predictions when environmental conditions vary.

C. Performance Comparison

To evaluate generalization under distribution shifts, the proposed method is compared with representative crack segmentation models, including AttentionCrack [12], CrackFormerII [13], LiteFusionNet [14], and OUR-Net [15], within a unified framework. All models are trained on the CamCrack789 training set with identical settings and evaluated on the cross-domain datasets (Crack500 and CFD).

Table II shows that most models suffer substantial degradation on cross-domain datasets, revealing sensitivity to variations in imaging conditions. In contrast, the proposed method exhibits the smallest performance drop and achieves the best overall results. On Crack500 and CFD, it attains F1-scores of 0.6417 and 0.8186, outperforming all competing models.

TABLE II
PERFORMANCE COMPARISON ACROSS TEST DATASETS

Model	Dataset	Precision	Recall	F1	IoU	Params	FLOPs
Baseline	Crack500	0.5816	0.5390	0.5595	0.4101	1.63M	2.30G
	CFD	0.7973	0.7743	0.7856	0.6470		
Ours	Crack500	0.6479	0.6357	0.6417	0.4724	1.63M	2.30G
	CFD	0.8105	0.8270	0.8186	0.6930		
AttentionCrack	Crack500	0.7534	0.3729	0.4989	0.3324	23.47M	35.08G
	CFD	0.6583	0.6336	0.6457	0.4768		
CrackFormerII	Crack500	0.5656	0.5829	0.5741	0.4027	4.55M	20.46G
	CFD	0.5769	0.6959	0.6308	0.4608		
LiteFusionNet	Crack500	0.5912	0.5176	0.5520	0.3812	3.95M	1.08G
	CFD	0.6493	0.7046	0.6759	0.5104		
OUR-Net	Crack500	0.7513	0.3848	0.5090	0.3414	24.69M	19.20G
	CFD	0.6529	0.6361	0.6444	0.4754		

To further illustrate cross-domain robustness, Fig. 4 shows representative results on CFD, Crack500, and combined scenarios. Under significant appearance shifts such as texture variations and background interference, several methods exhibit false negatives or discontinuous segmentation, whereas the proposed method maintains clearer structural integrity and reduced spurious responses.

These improvements are obtained without increasing model complexity. The proposed method maintains the same parameter count and FLOPs as the baseline (1.63M parameters and 2.30G FLOPs), while significantly outperforming much larger models, demonstrating its suitability for lightweight deployment.

To analyze the effect of the proposed channel attention mechanism, it is replaced with CBAM [16], NAM [17], ECA [18], and SE [19] under identical configurations. As reported in Table III, alternative modules provide limited or unstable improvements under cross-domain conditions, whereas CAM consistently achieves superior performance across datasets.

To further evaluate the domain generalization capability of the proposed method, we conduct bidirectional cross-domain experiments between CamCrack789 and Crack500. In addition, representative domain generalization methods,

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT ATTENTION MODULES

Method	Dataset	Precision	Recall	F1	IoU
CAM	CamCrack789	0.9173	0.9280	0.9226	0.8564
	Crack500	0.6479	0.6357	0.6417	0.4724
	CFD	0.8105	0.8270	0.8186	0.6930
CBAM	CamCrack789	0.8783	0.8847	0.8815	0.7881
	Crack500	0.2511	0.5791	0.3503	0.2123
	CFD	0.6529	0.6449	0.6489	0.4803
NAM	CamCrack789	0.9231	0.9140	0.9185	0.8493
	Crack500	0.7322	0.5154	0.6050	0.4337
	CFD	0.7628	0.7982	0.7801	0.6395
ECA	CamCrack789	0.9138	0.9309	0.9223	0.8558
	Crack500	0.6553	0.5596	0.6037	0.4323
	CFD	0.7792	0.7859	0.7825	0.6427
SE	CamCrack789	0.9237	0.9119	0.9177	0.8480
	Crack500	0.6534	0.5933	0.6219	0.4513
	CFD	0.7977	0.7881	0.7929	0.6568

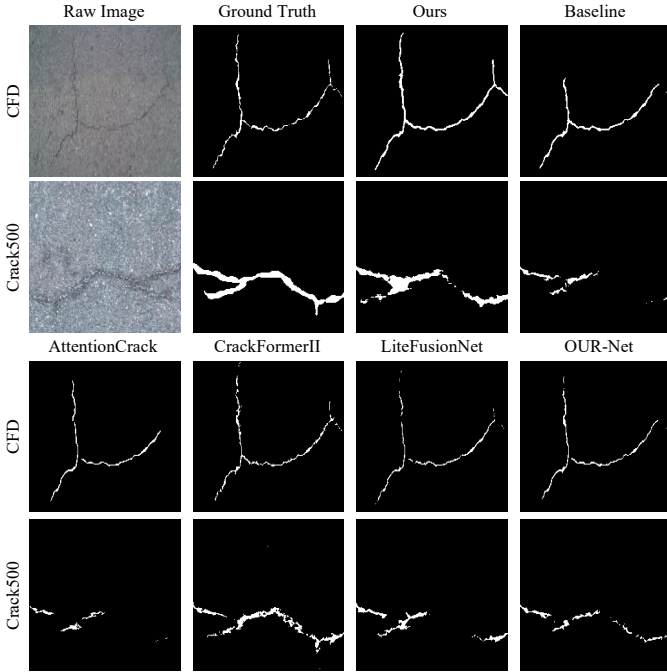


Fig. 4. Qualitative comparison of segmentation results on cross-domain datasets.

including SAN+SAW [20] and CIRL [4], are introduced for comparison to enhance the comprehensiveness and credibility of the evaluation.

As shown in Table IV, the proposed method consistently outperforms the baseline in both transfer directions, demonstrating its effectiveness in mitigating domain shifts. Compared with SAN+SAW, the proposed method exhibits more stable and balanced performance across different transfer directions. In contrast, CIRL shows relatively inferior performance in this task, indicating its limited applicability to fine-grained crack segmentation.

Overall, the results indicate that the proposed method

TABLE IV
CROSS-DOMAIN PERFORMANCE COMPARISON UNDER DIFFERENT SOURCE-TO-TARGET SETTINGS

Method	Source $\rightarrow$ Target	Precision	Recall	F1	IoU
Baseline	CamCrack789 $\rightarrow$ Crack500	0.5816	0.5390	0.5595	0.4101
	Crack500 $\rightarrow$ CamCrack789	0.6102	0.5233	0.5634	0.3922
Ours	CamCrack789 $\rightarrow$ Crack500	0.6479	0.6357	0.6417	0.4724
	Crack500 $\rightarrow$ CamCrack789	0.6269	0.6431	0.6349	0.4651
SAN+SAW	CamCrack789 $\rightarrow$ Crack500	0.6574	0.5665	0.6086	0.4374
	Crack500 $\rightarrow$ CamCrack789	0.5184	0.7770	0.6219	0.4513
CIRL	CamCrack789 $\rightarrow$ Crack500	0.5209	0.4358	0.4746	0.3111
	Crack500 $\rightarrow$ CamCrack789	0.5762	0.6184	0.5965	0.4251

improves robustness to distribution shifts and enables more stable crack segmentation across diverse environments and bidirectional cross-domain settings.

D. Real-World Deployment

To validate practical applicability, the proposed method was deployed on an embedded inspection platform consisting of a Jetson Xavier NX, an HD camera, and a Tracer 2.0 mobile robot base (See Fig. 5). The system processes road images onboard in real time without external computation.



Fig. 5. Real-world deployment platform and road crack segmentation results on the mobile robot.

Field experiments were conducted in an outdoor environment in Longhua District, Shenzhen. Input RGB images were resized to 480×320 pixels to balance accuracy and efficiency. The system achieved an average inference speed of 22.7 FPS, demonstrating real-time capability on resource-constrained embedded hardware.

These results confirm that the proposed lightweight model can operate reliably under real-world conditions and is suitable for deployment on autonomous inspection robots and other edge platforms for practical road monitoring tasks.

E. Failure Case

Although CAM improves robustness under diverse perturbations and cross-domain settings with negligible additional computational cost, some false positives still appear in complex backgrounds. As shown in Fig. 6, coarse aggregate boundaries may be misclassified as cracks because they share similar thin, irregular, and high-contrast linear patterns with real cracks.

This limitation mainly stems from semantic ambiguity rather than distribution shift. CAM is effective at suppressing environment-related disturbances, such as illumination, shadow, blur, and texture variations, but is less effective in distinguishing true cracks from crack-like background structures through channel-wise modulation alone.

Future work will consider multi-scale context, topology-aware constraints, and material-aware priors to further reduce false positives in real-world scenes.

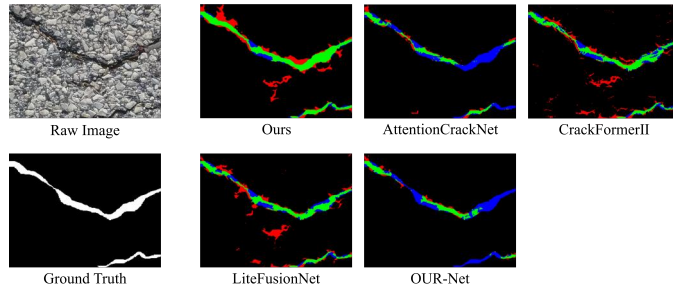


Fig. 6. Failure case involving concrete pavement with coarse aggregates.

V. CONCLUSION

Inspired by causal mechanisms, this work proposes a channel attention module to enhance the domain generalization ability of crack segmentation models. Experiments on Cam-Crack789, Crack500, and CFD demonstrate that the proposed module consistently improves robustness across varying conditions. When integrated into the decoder of RHACrackNet, it achieves superior generalization performance compared with state-of-the-art crack segmentation methods and yields greater gains than other common attention modules. Furthermore, the integrated model is deployed on an embedded platform within a mobile robot system, achieving real-time crack segmentation at 22.7 FPS in field tests, validating both effectiveness and practical feasibility. Future work will focus on further improving adaptability to more complex real-world road scenarios.

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